

FLEET RULE OF THE ROAD TEST APR 14

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test is completely scenario based.

You are OOW1 on a Type 45 Destroyer, sailing from Faslane to Portsmouth; you have the Forenoon and have just taken the ship from the NO on passing Cloch Point. The NO has confidently forecast light winds and good visibility for the entire passage.

1. As you proceed down the Firth of Clyde Channel, you note that a USN SSN is due to visit soon after your departure. The CO asks the SFT YO whether or not the SSN must comply with the provisions of the Rules with regard to the display of navigation lights. What do the Rules state? (3 marks) (R1e)

2. The SFT YO has prepared a pilotage run which takes the ship through Hunterston Channel. Whilst proceeding down this narrow channel the QM spots a small vessel engaged in fishing within the channel, at Green 15 range 5 cables.

a. What are the responsibilities of the respective vessels within this Narrow Channel in terms of who should avoid impeding whom, and why? (2 marks) (R9c)

b. Who is the Give Way Vessel between the fishing vessel and you if risk of collision exists and why? (2 marks) (R8f, iii)

3. You pass through Hunterston Channel and encounter a yacht race. Traffic is exceptionally heavy and it is becoming difficult to assess the situation and likelihood of risk of collision. What action shall you take? (2 marks) (R8e)

4. You take back the con and clear the yacht race. Contrary to the NO's brief you sight a thick fog bank approximately 1 NM away.

a. At what point shall you turn on your navigation lights and why? (2 marks) (R20)

b. Just as you enter the fog bank the MEO enters the bridge and requests to take the stbd engine to one hours notice in order to carry out some maintenance. Is this appropriate? (1 mark) (R19b)

c. You hear one prolonged blast ahead of the ship but are unsure as to the exact direction it came from. What action shall you take? (3 marks) (R19e)

d. Your foc'sle lookout reports a sound signal of one prolonged blast followed by two short blasts. What sort of vessel could this be? (3 marks) (R35c)

e. As you pass the entrance to Brodick Bay the Stbd Bridge wing lookout reports hearing a sound signal consisting of one short, one prolonged and one short sounded in succession. What precisely does this signify? (1 mark)(R35g)

5. This vessel passes clear and, still in restricted visibility, you next make out the following lights passing down your starboard side. What do they indicate? (4 marks) (R24a)

	W		
		R	W
		W	W
		R	
G	G		

6. Next you encounter a solitary vessel displaying the following lights. What do they signify? (3 marks) (R26b, i, ii and Annex II)

G
W

R
R

G

7. You are just about to go off watch when the SCC informs you that it has detected excessive vibration on the port shaft and as you are well ahead of the SOA the CO decides to go to anchor. What sound signals would you make at anchor in restricted visibility and what additional measures can you take to warn other vessels of your position? (3 marks) (R35g)

8. Whilst off watch the defect is fixed and you return for your next watch to find the ship back underway although visibility remains reduced. The CO asks for your assessment of safe speed. All radars appear to be working, list six considerations you should make regarding radar in determining your answer. (6 marks) (R6b)

9. Your next watch starts just as you round Land's End and with the visibility also improving the Flight become increasingly keen to disembark before the ship commences a busy period in Portsmouth so you take the ship to flying stations. As you give a green light you spot a minehunter engaged in clearance operations on a steady bearing, range 2 NM. Who has right of way? (1 mark) (ATP1E)

10. With the helo feet dry you fall out from flying stations and refocus on your SOA to Portsmouth. You sight a small vessel engaged in fishing at GREEN 30, range 3 miles. The bearing is steady and the range is closing. What action is required by the Rules? (1 mark)(R18)

11. The busy watch continues and you sight a coaster at RED 40, range 6 miles (60° on her stbd bow) . The range is closing and the bearing is moving very slowly right.

a. What action is required by the Rules? (1 mark)(R15)

b. You become concerned with the developing situation. Would a sound signal be appropriate and what should be sounded? (1 mark)(R34d)

c. When do the Rules allow the stand-on vessel to manoeuvre? (1 mark)(17a)

(1)

d. What specifically should be avoided if the stand-on vessel manoeuvres by altering course? (1 mark)(R17c)

12. You sight a vessel close in, apparently stopped off Fowey. How can you determine if the vessel is aground? (1 mark)(R30d)

13. As you alter to pass Dodman Point, you sight a small vessel displaying three shapes in a vertical line, the highest and lowest are ball shaped and the middle one a diamond.

a. How should you approach this vessel? (1 mark)(R27b/18)

b. List two operations which this small vessel could be engaged in. (2 marks)(R3g)

14. As you approach Rame Head in preparation for a boat transfer in Plymouth Sound to disembark the WEO, the lookout reports an open boat showing a rigid replica of a flag, which you identify as International Code flag "A". What does this signify? (1 mark)(R27e)

The BTX is complete and it's almost time to handover but light is starting to fade.

15. Can you exhibit navigation lights before sunset? Explain your answer. (1 mark)(R20c)

16. List three **visual** methods by which a small yacht without VHF could indicate distress by day. (Do **not** include light signals.) (3 marks)(R37/Annex IV)

Total _____ / 50

ANSWERS

1. As you proceed down the Firth of Clyde Channel, you note that a USN SSN is due to visit soon after your departure. The CO asks the SFT YO whether or not the SSN must comply with the provisions of the Rules with regard to the display of navigation lights. What do the Rules state? (3 marks) (R1e)

No. An SSN is a vessel of special construction/purpose and it is likely that her Government will have determined that the SSN cannot comply fully with this provision. She should however, comply with as closely as possible to the Rules in the respect of the provision of lights as her Government shall have determined in respect of that vessel.

2. The SFT YO has prepared a pilotage run which takes the ship through Hunterston Channel. Whilst proceeding down this narrow channel the QM spots a small vessel engaged in fishing within the channel, at Green 15 range 5 cables.

a. What are the responsibilities of the respective vessels within this Narrow Channel in terms of who should avoid impeding whom, and why? (2 marks) (R9c)

The vessel engaged in fishing shall avoid impeding your passage as a VEF shall not impede the passage of any other vessel navigating within the Narrow Channel or fairway

b. Who is the Give Way Vessel between the fishing vessel and you if risk of collision exists and why? (2 marks) (R8f, iii)

We are. A vessel the passage of which is not to be impeded remains fully obliged to comply with the Rules of this part when the two vessels are approaching one another so as to involve risk of collision.

3. You pass through Hunterston Channel and encounter a yacht race. Traffic is exceptionally heavy and it is becoming difficult to assess the situation and likelihood of risk of collision. What action shall you take? (2 marks) (R8e)

If necessary to avoid collision or allow more time to assess the situation, a vessel shall slacken her speed or take all way off by stopping or reversing her means of propulsion.

4. You take back the con and clear the yacht race. Contrary to the NO's brief you sight a thick fog bank approximately 1 NM away.

a. At what point shall you turn on your navigation lights and why? (2 marks) (R20)

Navigation lights should be turned on at this point. The Rule governing conduct of vessels in restricted visibility applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.

b. Just as you enter the fog bank the MEO enters the bridge and requests to take the port shaft to one hours notice in order to carry out some maintenance. Is this appropriate? (1 mark) (R19b)

No. In restricted visibility a power-driven vessel shall have her engines ready for immediate manoeuvre.

c. You hear one prolonged blast ahead of the ship but are unsure as to the exact direction it came from. What action shall you take? (3 marks) (R19e)

Reduce your speed to the minimum at which you can keep on course. If necessary take all way off and in any event navigate with extreme caution until danger of collision is over.

d. Your foc'sle lookout reports a sound signal of one prolonged blast followed by two short blasts. What sort of vessel could this be? (3 marks) (R35c)

NUC	Sailing
RAM	CBD
VEF	Towing or pushing

e. As you pass the entrance to Brodick Bay the Stbd Bridge wing lookout reports hearing a sound signal consisting of one short, one prolonged and one short sounded in succession. What precisely does this signify? (1 mark)(R35g)

A vessel at anchor may in addition (to other sound signals) sound three blasts in succession, namely one short, one prolonged, one short, to give warning of her position and of the possibility of collision to an approaching vessel.

5. This vessel passes clear and, still in restricted visibility, you next make out the following lights passing down your starboard side. What do they indicate? (4 marks) (R24a)

	W		
		R	W
		W	W
		R	
G	G		

A vessel engaged in towing seen from the starboard side who is severely restricted in its ability to manoeuvre. The towing vessel is less than 50m in length while the length of the tow is over 200m in length.

6. Next you encounter a solitary vessel displaying the following lights. What do they signify? (3 marks) (R26b, i, ii and Annex II)

	G	
	W	
R		
R		
		G

A vessel engaged in trawling (under 50m in length) whose nets/trawls have come fast on an obstruction (viewing a stbd aspect).

7. You are just about to go off watch when the SCC informs you that it has detected excessive vibration on the port shaft (the MEO did warn you in Q4) and as you are well ahead of the SOA the CO decides to go to anchor. What sound signals would you make at anchor in restricted visibility and what additional measures can you take to warn other vessels of your position? (3 marks) (R35g)

5 seconds ringing of the bell in the forepart of the ship at intervals of not more than one minute followed immediately by rapidly sounding a gong for 5 seconds in the after part of the ship.

In addition sound three blasts in succession, namely one short, one prolonged and one short blast, to give warning of position.

8. Whilst off watch the defect is fixed and you return for your next watch to find the ship back underway although visibility remains reduced. The CO asks for your assessment of safe speed. All radars appear to be working, list six considerations you should make regarding radar in determining your answer. (6 marks) (R6b)

- i) The characteristics, efficiency and limitations of the radar equipment.**
- ii) Any constraints imposed by the radar range scale in use.**
- iii) The effect on radar detection of the sea state, weather and other sources of interference.**
- iv) The possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range.**
- v) The number, location and movement of vessels detected by radar.**
- vi) The more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.**

9. As you round Land's End the visibility improves and as usual the Flight is keen to disembark before the ship commences a busy period in Portsmouth so you take the ship to flying stations. As you give a green light you spot a minehunter engaged in clearance operations on a steady bearing, range 2 NM. Who has right of way? (1 mark) (ATP10D)

The minehunter

10. With the helo feet dry you fall out from flying stations and refocus on your SOA to Portsmouth. You sight a small vessel engaged in fishing at GREEN 30, range 3 miles. The bearing is steady and the range is closing. What action is required by the Rules? (1 mark)(R18)

A power-driven vessel underway shall keep out of the way of a vessel engaged in fishing.

11. The busy watch continues and you sight a coaster at RED 40, range 6 miles (60° on her stbd bow) . The range is closing and the bearing is moving very slowly right.

- a. What action is required by the Rules? (1 mark)(R15)

When two power-driven vessels are crossing so as to involve a risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

- b. You become concerned with the developing situation. Would a sound signal be appropriate and what should be sounded? (1 mark)(R34d)

(1)

When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the

vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle.

c. When do the Rules allow the stand-on vessel to manoeuvre? (1 mark)(17a)

(1)

As soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

d. What specifically should be avoided if the stand-on vessel manoeuvres by altering course? (1 mark)(R17c)

A power-driven vessel which takes action in a crossing situation in accordance with sub-paragraph (a)(ii) of this Rule to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.

12. You sight a vessel close in, apparently stopped off Fowey. How can you determine if the vessel is aground? (1 mark)(R30d)

A vessel aground shall exhibit the lights prescribed in paragraph (a) or (b) of this Rule and in addition, where they can best be seen: three balls in a vertical line.

13. As you alter to pass Dodman Point, you sight a small vessel displaying three shapes in a vertical line, the highest and lowest are ball shaped and the middle one a diamond.

a. How should you approach this vessel? (1 mark)(R27b/18)

A power-driven vessel underway shall keep out of the way of a vessel restricted in her ability to manoeuvre.

b. List two operations which this small vessel could be engaged in. (2 marks)(R3g)

The term vessels restricted in their ability to manoeuvre shall include but not limited to: (any two out of the three options) a vessel servicing a navigation mark; a vessel engaged in surveying or underwater operations (diving). (Two choices.)

14. As you approach Rame Head in preparation for a boat transfer in Plymouth Sound to disembark the WEO, the lookout reports an open boat showing a rigid replica of a flag, which you identify as International Code flag "A". What does this signify? (1 mark)(R27e)

Whenever the size of a vessel engaged in diving operations makes it impracticable to exhibit all lights and shapes prescribed in paragraph (d) of this Rule, the following shall be exhibited: a rigid replica of the International Code flag "A" not less than 1 m in height. Measures shall be taken to ensure its all-round visibility.

The BTX is complete and it's almost time to handover but light is starting to fade.

15. Can you exhibit navigation lights before sunset? Explain your answer. (1 mark)(R20c)

Yes. The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility and may be exhibited in all other circumstances deemed necessary.

16. List three **visual** methods by which a small yacht without VHF could indicate distress by day. (Do **not** include light signals.) (3 marks)(R37/Annex IV)

When a vessel is in distress and requires assistance she shall use or exhibit signals described in annex IV of these Regulations.

Three of the following:

Rockets or shells throwing red stars fired one at a time at short intervals;

A signal light consisting of a square flag having above or below it a ball or anything resembling a ball;

A rocket parachute flare or a hand-flare showing a red light.

A smoke signal giving off orange coloured smoke;

Slowly and repeatedly raising and lowering arms outstretched to each side.

Total _____ / 50